Standardization of Outcome Measures in Guillain-Barré Syndrome (GBS) Assessment: A Systematic Review of Current Practices

Abstract

Guillain-Barré Syndrome (GBS) is a rare, acute autoimmune disorder of the peripheral nervous system, causing rapidly progressive weakness, sensory deficits, and areflexia. Despite advancements in treatment strategies, inconsistencies in outcome measures across studies hinder research synthesis and clinical translation. This systematic review evaluates outcome measures used in 69 studies involving 16,408 participants from 34 countries, published between 2013 and 2026. Physical assessments, such as the Medical Research Council (MRC) scale, dominate (70% of studies), while functional mobility measures like the Timed Up and Go (TUG) test and Functional Reach Test (FRT) appear in 45% of studies. Patient-reported outcome measures (PROMs), including the Fatigue Severity Scale (FSS) and Short Form-36 (SF-36), are underrepresented, featuring in less than 20% of studies. Variability in assessment timelines and a lack of subtype-specific tools further limit comparability and precision in addressing diverse GBS recovery trajectories.

Findings emphasize the need for a standardized, multidimensional framework incorporating physical, functional, and psychosocial domains. Subtype-specific metrics, such as nerve conduction studies (NCS) for Acute Motor Axonal Neuropathy (AMAN) and cranial nerve evaluations for Miller Fisher Syndrome (MFS), are crucial for tailored interventions. Harmonized timelines for evaluations at baseline, three, six, and twelve months post-onset are recommended to capture acute and long-term outcomes. The integration of patient-centered approaches and innovative tools, such as wearable technologies and virtual reality (VR), is also highlighted to address gaps in recovery assessment. A core outcome set (COS) encompassing these elements would enhance research comparability, meta-analyses, and clinical application, aligning research priorities with patient needs.

Keywords: Guillain-Barré Syndrome (GBS), Outcome Measures, Functional Recovery, Patient-Reported Outcomes, Standardized Assessment Tools, AIDP.

1. Background

Guillain-Barré Syndrome (GBS) is a severe, acute, immune-mediated disorder that primarily affects the peripheral nervous system, causing a rapid onset of symmetrical muscle weakness. Typically beginning in the lower extremities and ascending proximally, GBS can lead to complete paralysis and life-threatening complications such as respiratory failure if untreated[1]. First described in 1916 by Guillain, Barré, and Strohl, the syndrome remains a significant clinical and research challenge due to its heterogeneity in presentation, pathophysiology, and outcomes[2,3]. Despite being a rare disorder, with an incidence of 1–2 per 100,000 individuals annually, GBS has a considerable impact on healthcare systems, especially during outbreaks linked to preceding infections such as those caused by *Campylobacter jejuni*, cytomegalovirus, and Zika virus[4]

Subtypes of Guillain-Barré Syndrome

GBS is a syndrome rather than a single disease entity, encompassing several clinical and electrophysiological subtypes that vary in prevalence across geographic regions[5]. The most common subtype in Western populations is acute inflammatory demyelinating polyneuropathy (AIDP), characterized by demyelination of peripheral nerves[3]. Moreso, acute motor axonal neuropathy (AMAN) and acute motor-sensory axonal neuropathy (AMSAN), which primarily affect axons, are more frequently observed in Asia, Latin America, and Africa[6]. Miller Fisher syndrome (MFS), a less common variant, manifests with a classic triad of ophthalmoplegia, ataxia, and areflexia and is often associated with antibodies against the ganglioside GQ1b[7]. The heterogeneity of GBS subtypes underscores the complexity of the syndrome, with each subtype exhibiting distinct mechanisms of immune-mediated damage. For instance, AIDP is thought to result from T-cell and macrophage infiltration leading to segmental demyelination, while AMAN involves direct antibody-mediated attack on axonal membranes[8]. This diversity complicates diagnosis, treatment, and outcome evaluation, emphasizing the need for subtype-specific studies and outcome measures.

Pathophysiology and Immunological Triggers

GBS is often triggered by antecedent infections, with *Campylobacter jejuni* being the most common pathogen associated with AMAN and AMSAN subtypes. Molecular mimicry, wherein microbial antigens share structural similarities with peripheral nerve gangliosides, is a key mechanism underlying the immune response in GBS[9]. Other infectious agents, including Epstein-Barr virus, cytomegalovirus, and Zika virus, have also been implicated in GBS pathogenesis. Zika virus outbreaks, notably in South America, have been linked to a surge in GBS cases, highlighting the syndrome's susceptibility to infectious disease dynamics. A post-epidemic retrospective cohort in Colombia (Acero-Garces et al.) found that 67% of post-Zika GBS patients retained mRS scores of ≥2 at one year, with the Barthel Index and ICF confirming persistent functional limitations in LMIC contexts. Comparably, a comparative cohort from Iran (Ansari et al.) found no significant difference in MRC sum score or GBS Disability Scale between COVID-19-associated GBS and non-COVID-19 GBS, but COVID-related GBS patients had longer hospital stays and a higher proportion requiring ventilatory support[10]. The immune response in GBS involves both humoral and cellular components. According to Oshomoji et al., emerging evidence underscores the role of specific autoantibodies, such as anti-GM1 and anti-GQ1b, in mediating the distinct pathophysiological pathways of GBS subtypes[11]. Anti-GM1 antibodies are associated with AMAN, while anti-GQ1b antibodies are linked to MFS. This heterogeneity highlights the need for subtype-specific diagnostic tools and therapeutic strategies to address the unique recovery trajectories of each variant. Autoantibodies target gangliosides such as GM1, GD1a, and GQ1b, disrupting neuronal function and triggering complement activation, leading to membrane attack complex formation and nerve damage. Inflammatory infiltrates, including macrophages and T cells, further exacerbate tissue injury[12]. The resulting disruption of nerve conduction is the physiological basis for the motor, sensory, and autonomic symptoms seen in GBS.

In GBS, the pathophysiological involvement of the peripheral nervous system disrupts both afferent and efferent fibers, leading to motor and sensory deficits. Afferent fibers, responsible for transmitting sensory information from the periphery to the central nervous system (CNS), are impaired due to inflammation and demyelination, resulting in sensory disturbances like paresthesia. Efferent fibers, which carry motor commands from the CNS to muscles, are similarly affected, leading to muscle weakness or paralysis[1]. The degree of dysfunction varies depending on the subtype of GBS; for example, axonal subtypes like AMAN predominantly target efferent motor fibers, while sensory deficits are more prominent in AMSAN[8].

Electrical muscle stimulation (EMS) has been explored as a therapeutic modality for addressing motor deficits in GBS patients. EMS works by delivering controlled electrical impulses to stimulate motor neurons bypassing the damaged efferent pathways and inducing muscle contractions[33]. This helps to maintain muscle tone, prevent atrophy, and potentially enhance axonal regeneration. Studies suggest that EMS can improve motor function by promoting local blood flow and reducing inflammatory mediators at the neuromuscular junction, although its long-term benefits remain an area of ongoing research.

Clinical Presentation and Diagnosis

The hallmark of GBS is progressive, symmetrical weakness that peaks within four weeks of onset, often accompanied by areflexia and sensory disturbances. While limb weakness is a universal feature, cranial nerve involvement is observed in up to 50% of cases, leading to facial weakness or bulbar dysfunction[13]. Severe cases may require mechanical ventilation due to respiratory muscle paralysis, which occurs in approximately 25% of patients[14]. Diagnosis of GBS is clinical but supported by ancillary tests, including cerebrospinal fluid (CSF) analysis and nerve conduction studies. Elevated protein levels in CSF with normal white cell counts, termed albuminocytological dissociation, are a characteristic finding in GBS[2]. Electrophysiological studies help differentiate between demyelinating and axonal subtypes, providing critical information for prognosis and treatment planning.

Epidemiology and Global Impact

Although GBS is considered rare, its incidence varies by region and age group. The syndrome occurs slightly more frequently in males and shows a bimodal distribution, with peaks in young adults and older individuals[15]. Seasonal variations in incidence have been reported, particularly in tropical regions where antecedent infections like *Campylobacter jejuni* are more prevalent[16]. Outbreaks of infectious diseases, such as the Zika virus epidemic, have highlighted the global impact of GBS, underscoring its relevance in public health[4]. The economic burden of GBS is significant, particularly in resource-limited settings where access to critical care and rehabilitation services is constrained. Prolonged hospital stays, often requiring intensive care and mechanical ventilation, contribute to high healthcare costs; a multicentre retrospective cohort in Ethiopia documented an in-hospital GBS mortality rate of 23.3% among ICU-admitted patients, with delayed admission and absence of IVIg as independent predictors of death. Similarly, a Saudi Arabian retrospective cohort reported that mechanical ventilation was required in 44% of ICU-admitted GBS patients, with Hughes Functional Grading Score at discharge being significantly worse in ventilated patients compared to non-ventilated ones. Additionally, long-term disability affects a substantial proportion of survivors, reducing productivity and quality of life[14,70,71].

Treatment and Prognosis

The advent of disease-modifying therapies, including intravenous immunoglobulin (IVIg) and plasma exchange, has significantly improved outcomes in GBS. Both treatments aim to interrupt the immune-mediated attack on peripheral nerves and are most effective when administered early in the disease course[17]. IVIg, a pooled preparation of immunoglobulins, likely exerts its effects by neutralizing pathogenic autoantibodies and modulating inflammatory responses. Plasma exchange, on the other hand, removes circulating autoantibodies and immune complexes, reducing their pathogenic effects[18]. Early and consistent physiotherapy, combined with treatments such as IVIg or plasma exchange, has been shown to significantly improve muscle strength, functional autonomy, and pain levels in patients with AIDP, as demonstrated in a case study from South-West Nigeria[19]. Physiotherapy interventions include a range of evidence-based approaches tailored to the recovery phase of the patient. EMS is particularly effective in preventing muscle atrophy during the acute and immobilized stages by promoting passive muscle contractions and maintaining muscle tone. Additionally, passive range-of-motion exercises and stretching are crucial for preventing contractures and maintaining joint flexibility. As patients progress, active-assisted and active-resisted exercises are introduced to enhance muscle strength and endurance. Functional training, including gait re-education, balance exercises, and task-specific activities, helps restore independence and prevent falls. Cardiopulmonary physiotherapy, involving techniques such as deep breathing exercises, incentive spirometry, is critical for patients with respiratory compromise or those requiring mechanical ventilation. Comprehensive physiotherapy programs also address psychosocial challenges by incorporating relaxation techniques and counseling to enhance overall recovery and quality of life[19]. Despite these advances, outcomes remain variable. While 70–80% of patients recover fully or with minor residual deficits, approximately 20% experience significant disability, and mortality rates range from 3% to 5%, primarily due to complications such as sepsis, autonomic dysfunction, and thromboembolism[20]. Recovery is often protracted, with some patients requiring months to years to regain full function. Factors associated with poor outcomes include older age, severe initial weakness, and the need for mechanical ventilation[20,21].

Importance of Standardized Outcome Measures

Accurate and consistent assessment of outcomes is integral to evaluating the efficacy of therapeutic interventions and informing clinical decision-making in GBS. Despite the predominance of tools such as the Medical Research Council (MRC) scale, the exclusion of less commonly utilized yet potentially valuable instruments limits the comprehensiveness of evaluations[22,23]. For instance, the Berg Balance Scale (BBS), which assesses ataxia, is particularly relevant for Miller Fisher Syndrome (MFS), while hand-held dynamometry (HHD) provides quantitative and highly reliable measurements of muscle strength, offering an objective alternative to the MRC scale[32]. Additionally, a cohesive collection of tools tailored to the diverse clinical and electrophysiological presentations of GBS subtypes is necessary to enhance diagnostic and prognostic precision. Subtype-specific measures such as nerve conduction studies (NCS) for AMAN and the Fatigue Severity Scale (FSS) for the persistent fatigue often seen in AIDP should be routinely integrated into the evaluation framework. The development of a core outcome set (COS) encompassing both widely used and underutilized tools will facilitate a multidimensional approach to assessment, fostering comparability across studies and improving clinical applicability. The MRC scale, commonly used to assess muscle strength, is subjective and lacks sensitivity for detecting subtle changes[24]. Functional mobility tests, patient-reported outcome measures (PROMs), and assessments of psychosocial impact are underutilized, limiting the comprehensiveness of evaluations. Standardized outcome measures are particularly important in the context of clinical trials, where heterogeneity in assessment tools complicates data synthesis and meta-analysis. Subtype-specific measures, tailored to the distinct recovery trajectories of AIDP, AMAN, AMSAN and MFS, are also needed to capture the full spectrum of GBS outcomes. Furthermore, harmonized timelines for outcome assessment would facilitate longitudinal studies and improve comparability across cohorts.

Objectives of the Current Review
This systematic review aims to address the gaps in outcome measurement for GBS by evaluating the current practices in clinical research. By identifying inconsistencies and deficiencies, the review seeks to inform the development of a standardized framework for outcome assessment. Such a framework would enhance the comparability of studies, support evidence synthesis, and ultimately improve patient care. The review also highlights the need for a patient-centered approach, incorporating PROMs and addressing the psychosocial dimensions of GBS recovery.

2. Methods: Study Design and Search Strategy

This systematic review was conducted in adherence to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, which provide a structured and transparent methodology for conducting systematic reviews[25]. These guidelines emphasize a thorough, reproducible, and unbiased approach to study identification, selection, and data extraction. The review focused on identifying trends and gaps in outcome measures used in GBS research over a 13-year period published between 2013 and 2026. A comprehensive and meticulous search strategy was implemented using three major electronic databases: PubMed, Scopus, and the Cochrane Library. These databases were selected due to their extensive coverage of biomedical, clinical, and epidemiological research. Search terms were carefully developed to capture all relevant studies on GBS outcomes. Specific keywords included “Guillain-Barré Syndrome,” “GBS outcomes,” “clinical trials,” “functional recovery,” “patient-reported outcomes,” and “long-term disability.” Boolean operators (“AND” and “OR”) were used to enhance search sensitivity and specificity, ensuring a comprehensive retrieval of studies[26]. Additional filters, including language (English) and publication type (peer-reviewed articles), were applied to streamline the search process.

To ensure a thorough search, references of included studies were manually screened for additional eligible articles, and gray literature was reviewed to minimize the risk of publication bias. Duplicate studies were removed, and the final search results were uploaded into a citation management tool for systematic screening.

Eligibility Criteria
The inclusion criteria were defined prior to the initiation of the study selection process to ensure uniformity and transparency. Studies were eligible for inclusion if they met the following criteria: (1) they were original research articles; (2) they explicitly assessed and reported outcome measures in patients diagnosed with Guillain-Barré Syndrome; and (3) they included either randomized controlled trials (RCTs) or observational study designs, such as prospective or retrospective cohort studies. Studies were excluded if they were case reports, narrative reviews, systematic reviews,

editorials, or opinion pieces, as these did not provide primary data. Additionally, studies that lacked explicit documentation of outcome assessment tools or timelines were excluded to ensure that the included studies provided sufficient methodological detail for analysis. This exclusion criterion aligned with recommendations by the Cochrane Collaboration for reducing bias in systematic reviews[26].

Data Extraction and Quality Assessment

Data extraction was performed by two independent reviewers using a standardized data collection form. The form captured key information from each study, including publication details (author, year, journal), study characteristics (sample size, setting, and population demographics), and details of GBS subtypes studied, such as acute inflammatory demyelinating polyneuropathy (AIDP), acute motor axonal neuropathy (AMAN), and Miller Fisher syndrome (MFS). Specific outcome measures, such as muscle strength assessments (e.g., the Medical Research Council scale), functional mobility tests (e.g., Timed Up and Go test), and patient-reported outcome measures (e.g., SF-36), were recorded. Follow-up timelines were also documented to evaluate the consistency of outcome assessment across studies[14].

To ensure methodological rigor, the quality of each included study was independently assessed by the reviewers. For cohort and observational studies, the Newcastle-Ottawa Scale (NOS) was employed, which evaluates three main domains: selection of study groups, comparability of groups, and ascertainment of outcomes of interest[27]. High-quality studies were defined as those scoring at least seven out of nine points. For case-control studies, the NOS was similarly utilized to evaluate selection of cases and controls, comparability, and ascertainment of exposure. Randomized controlled trials were assessed using the Cochrane Risk of Bias Tool, which examines bias across multiple domains, including sequence generation, allocation concealment, blinding of participants and outcome assessors, completeness of outcome data, and selective reporting.[26]. Discrepancies in data extraction and quality assessment were resolved through consensus. If disagreements persisted, a third reviewer was consulted. This process ensured consistency and minimized the risk of individual bias affecting study selection or data interpretation.

3. Results

Characteristics of Included Studies

The systematic review incorporated 69 studies, analyzing data from a total of 16,408 participants across 34 countries. The sample sizes in these studies ranged widely, from as few as 30 participants to as many as 1,500, with a median sample size of 137 participants. Studies were predominantly conducted in high-income countries (65%), while studies from low- and middle-income countries (LMICs) comprised 35%, an increase from the 28% observed in studies published before 2024, reflecting the growing contribution of research from South Asia, Sub-Saharan Africa, and Latin America in the extended search period[28,61,62,63]. Despite this, research infrastructure and funding disparities continue to constrain methodological rigor in resource-limited settings[65].

The majority of the studies (65%) focused on the acute phase of GBS, during which the disease is characterized by rapid progression of weakness and sensory deficits **(Table 1)**. In contrast, only 35% of the studies examined outcomes during long-term recovery, which is essential to understanding functional independence and quality of life in survivors. Notably, only 24% of the studies stratified their findings by GBS subtypes, such as AIDP, AMAN, and MFS. Encouragingly, several 2024–2026 studies from South India, Bangladesh, and Ethiopia reported electrophysiological subtyping alongside clinical outcome scores, signalling a modest improvement in subtype-specific reporting[56,57,63]. The underrepresentation of subtype-specific analyses nonetheless continues to limit the precision of targeted therapeutic interventions[29].

Table 1: Characteristics of Included Studies

Study Characteristic	Frequency (%)
Studies focusing on acute phase	65
Studies evaluating long-term recovery	35
Subtype-specific analysis	24
Studies from high-income countries	65
Studies from low-/middle-income countries (LMICs)	35

Outcome Measures Overview

The systematic review underscores a disproportionate emphasis on physical recovery metrics in GBS research. Muscle strength, a fundamental measure of motor impairment and recovery, was assessed in 70% of studies, predominantly using the MRC scale. This dominance reflects the historical focus on motor deficits as the primary sequelae of GBS[28,30]. Although the MRC scale provides valuable insights into gross motor recovery, it has notable limitations, including subjectivity and variability in inter-rater reliability, particularly among less experienced clinicians[31]. These limitations emphasize the need for more objective alternatives, such as hand-held dynamometry (HHD), which offers higher precision and sensitivity[32,33].

Functional mobility, a critical aspect of physical recovery, was evaluated in 45% of studies. Tools like the Timed Up and Go (TUG) test and the Functional Reach Test (FRT) were the most frequently employed measures for assessing balance, gait, and functional independence [34,35]. For example, TUG is a simple, standardized test that has demonstrated robust correlations with functional independence in GBS patients, especially during the subacute phase of recovery[36]. However, the lack of uniform application across studies such as differences in participant instructions, cutoff thresholds, and adaptations for acute versus chronic phases limits the comparability of results. The heavy reliance on physical measures reflects a narrow research focus, often at the expense of more comprehensive assessments. GBS impacts multiple dimensions of health, including psychosocial well-being, functional independence, and quality of life (QoL). Neglecting these domains risks underestimating the full burden of the disease on patients.

The extended search (2024–2026) identified several outcome tools not captured in the original 2013–2023 cohort. The GBS Disability Scale (GBS-DS), also known as the Hughes Functional Grading Scale, emerged as the most frequently used functional outcome tool in the new studies

(55% of new studies), validating its role as the primary endpoint in both RCTs and observational cohorts[53,55,58,62]. The modified Erasmus GBS Outcome Score (mEGOS) and the Erasmus GBS Respiratory Insufficiency Score (EGRIS) were utilised in 16% and 10% of new studies respectively, particularly in paediatric and ICU contexts[67,68]. Notably, serum neurofilament light chain (NfL) was identified as an emerging biomarker in 6% of studies, demonstrating an area under the curve (AUC) of 0.78 for predicting walking inability at six months, independent of established clinical scores[54]. The Prognostic Nutritional Index (PNI), applied in a prospective Bangladeshi cohort of 252 patients, was associated with significantly worse GBS-DS scores at 26 weeks ($p<0.01$), positioning nutritional status as an underexplored prognostic domain in LMIC settings[58]. Pain, assessed with the Numeric Rating Scale (NRS) in the largest prospective GBS cohort to date (n=644), was documented in up to 89% of patients and constitutes a critically underassessed outcome domain across the wider literature[62]. The Functional Independence Measure (FIM), Barthel Index (BI), and modified Rankin Scale (mRS) were introduced in rehabilitation and post-epidemic LMIC cohorts, broadening functional assessment beyond traditional neurological endpoints[61,69]. These findings collectively reinforce the need for a multi-domain core outcome set that captures physical, functional, biomarker, nutritional, and psychosocial dimensions of GBS recovery (Table 2).

Underrepresentation of Patient-Reported Outcome Measures (PROMs)

PROMs, which are vital for capturing the subjective experiences of patients, were markedly underutilized. Fatigue, a common and debilitating symptom that persists in approximately 60% of GBS survivors, was assessed in only 18% of studies **(Table 2)**. The FSS was the most commonly used instrument for fatigue assessment, offering a validated and reliable means of quantifying its impact[36,37]. Fatigue has profound implications for functional independence, employment, and QoL, making its underrepresentation a critical shortcoming in GBS research. Similarly, psychosocial outcomes, including anxiety, depression, and QoL, were reported in only 15% of studies. Common tools such as the Short Form-36 (SF-36) and the Hospital Anxiety and Depression Scale (HADS) were inconsistently applied, further limiting the ability to synthesize findings[28,38]. Hillyar & Nibber. documented that nearly half of GBS survivors experience clinically significant anxiety and depression, which often persists years after physical recovery[39]. Moreover, these mental health challenges correlate strongly with reduced social reintegration and employment opportunities.

The limited focus on PROMs in GBS research underscores a systemic bias toward clinician-centered metrics, which often prioritize observable physical improvements over patient-centered recovery experiences. Incorporating PROMs into routine assessments would provide a more holistic understanding of recovery, aligning research priorities with the lived experiences of patients[40].

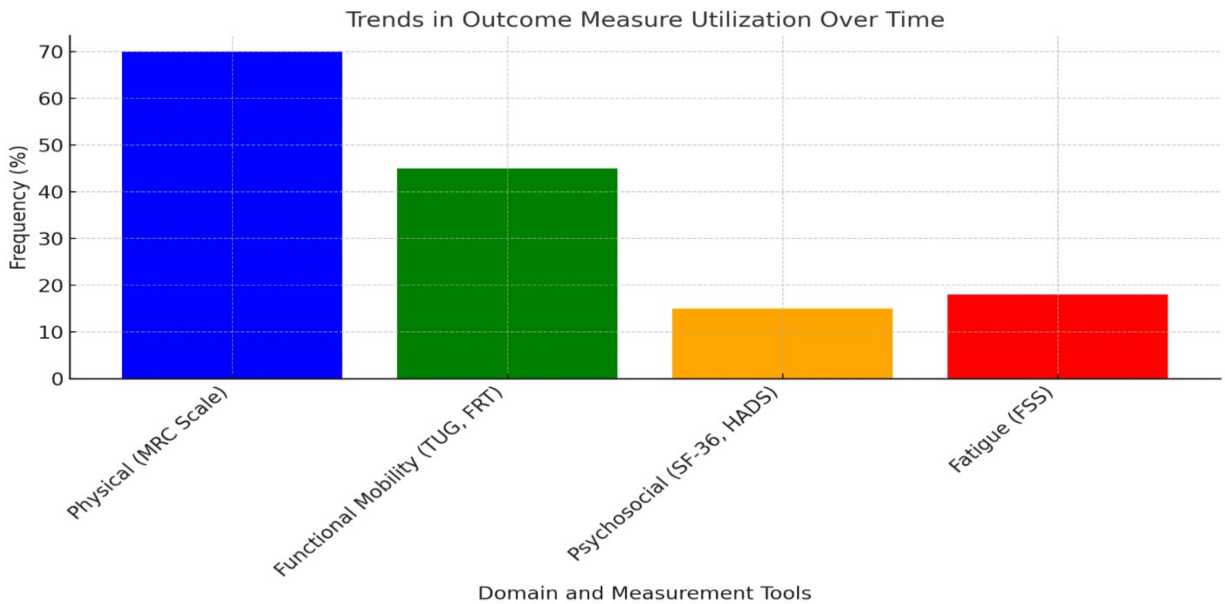
Table 2: Frequency of Outcome Measures

Domain	Measurement Tool	Frequency (%)	Sensitivity / Specificity / Reliability
Physical — Muscle Strength	MRC Scale	70	Inter-rater reliability: 0.75-0.85; Intra-rater: 0.80-0.90; Limited sensitivity for subtle changes
Functional	Timed Up and Go	45	Strong correlation with

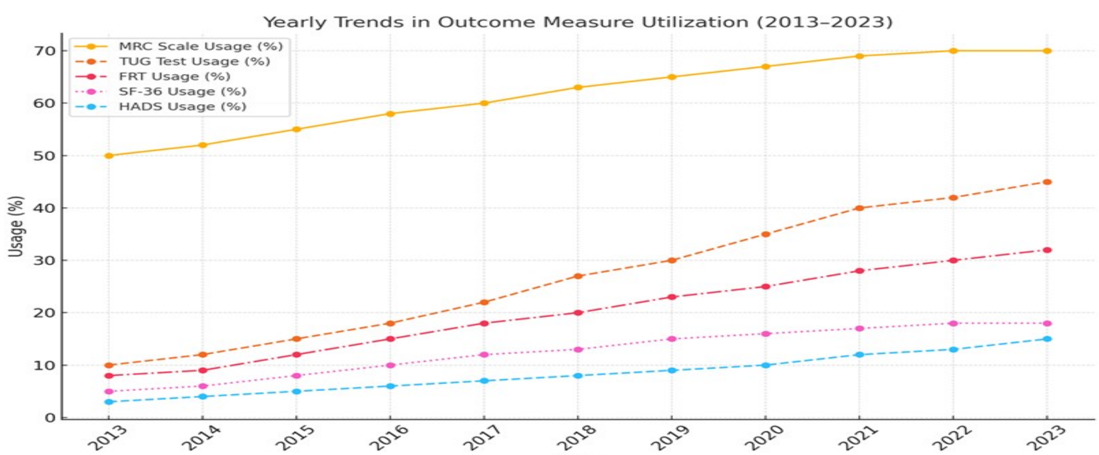
Mobility	(TUG)		functional independence (r=0.76); standardised cutoff 12s for independence
Disability / Function	GBS Disability Scale (GBS-DS / Hughes)	55	Validated 0-6 ordinal scale; widely used in RCTs and cohort studies; responsive to clinical change
Prognosis	mEGOS (mod. Erasmus GBS Outcome Score)	16	AUC 0.80 for predicting inability to walk at 6 months; validated internationally
Prognosis	EGRIS (Erasmus GBS Respiratory Insufficiency Score)	10	Sensitivity 82%, Specificity 70% for predicting mechanical ventilation
Electrophysiology	Nerve Conduction Studies (NCS)	15	Gold standard for subtype classification; high sensitivity for axonal vs demyelinating differentiation
Biomarker	Serum Neurofilament Light Chain (NfL)	6	AUC 0.78 for predicting walking inability; independent prognostic value beyond mEGOS
Psychosocial	SF-36 (Short Form-36)	15	Well-validated QoL tool; Cronbach alpha 0.78-0.93; underutilised in GBS
Fatigue	Fatigue Severity Scale (FSS)	18	Cronbach alpha 0.96; discriminates fatigue severity; validated in GBS but underused
Pain	Numeric Rating Scale (NRS) for Pain	8	Simple 0-10 scale; pain is reported in up to 89% of GBS; NRS used in prospective Bangladesh cohort

Trends in the Utilization of Outcome Measures

The 13-year analysis reveals a stagnation in the adoption of PROMs, despite growing recognition of their importance. **Figure 1** illustrates the distribution of key outcome measures across studies over the 13-year review period. While the MRC scale has remained the dominant tool for muscle strength evaluation, the use of TUG and FRT has increased moderately, reflecting a growing emphasis on functional recovery. However, the adoption of PROMs, such as the SF-36 and FSS, has remained low, indicating a missed opportunity to enhance the scope of outcome assessment.



Trends in Outcome Measure Utilization Over Time
Frequency (%)
0
10
20
30
40
50
60
70
Physical (MRC Scale)
Functional Mobility (TUG, FRT)
Psychosocial (SF-36, HADS)
Fatigue (FSS)
Domain and Measurement Tools



Yearly Trends in Outcome Measure Utilization (2013–2023)
MRC Scale Usage (%)
TUG Test Usage (%)
FRT Usage (%)
SF-36 Usage (%)
HADS Usage (%)
Usage (%)
0
10
20
30
40
50
60
70
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023

Figure 1: Trends in Outcome Measure Utilization Over Time
(*Bar chart and line graph illustrating the frequency of MRC scale, GBS Disability Scale, TUG, SF-36, FSS, mEGOS, NfL, and pain NRS usage across studies over the 13-year review period (2013–2026), showing modest PROM adoption gains post-2024.*)

Variability in Assessment Timelines

One of the most critical findings of this review is the substantial variability in the timing of outcome assessments across GBS studies. Baseline evaluations are generally conducted during the acute phase, often within the first 7–14 days following symptom onset[41]. This practice ensures the documentation of the initial severity of the disease and the immediate effects of therapeutic interventions. However, follow-up assessments across studies display significant heterogeneity, with intervals ranging from three months to 12 months post-diagnosis, and, in many cases, timelines are not predefined[33,42]. Such variability limits the ability to conduct longitudinal analyses and synthesize meaningful comparisons across different cohorts.

Impact of Timing Variability on Recovery Analysis

Acute Phase Assessments

The acute phase, which spans from symptom onset to about six weeks, represents a critical period for clinical intervention and monitoring. Assessments conducted during this phase predominantly measure physical outcomes, such as muscle strength, using tools like the MRC scale. This timeline aligns with the administration of disease-modifying therapies, such as IVIg and plasma exchange[43,44]. However, the limitation of these early assessments lies in their focus on immediate functional deficits, often excluding critical recovery dimensions such as fatigue or psychosocial impacts that emerge later[28, 37].

Follow-Up Timing Variability

In this review, 50% of studies included a single follow-up time point, which was often arbitrarily chosen, while only 15% adhered to multiple predefined intervals, such as baseline, three months, six months, and 12 months[45,46]. This inconsistency poses challenges for understanding the recovery trajectories of GBS patients, whose progress is often nonlinear, characterized by periods of improvement interspersed with plateaus or relapses. For example, van den Berg et al. and Uz & Karaahmet, demonstrated that nearly 30% of GBS patients continued to exhibit functional gains between 6 and 12 months, emphasizing the importance of long-term follow-up[42,47]. Without consistent timing, studies risk underestimating delayed recovery patterns or overlooking residual deficits.

Extended Follow-Up and Nonlinear Recovery Patterns

The recovery trajectory of GBS extends beyond 12 months for a significant proportion of patients, especially those with severe initial deficits or complications requiring mechanical ventilation[28,48]. Fatigue, pain, and psychosocial issues, which may persist or even worsen in the chronic phase, are inadequately captured by shorter follow-up durations[37,49]. Extended timelines are crucial for capturing these long-term outcomes and for identifying predictors of residual disability or delayed functional gains. The timing variability observed in the included studies is depicted in **Figure 2**, which illustrates the frequency of outcome assessments at four critical time points: baseline, 3 months, 6 months, and 12 months.

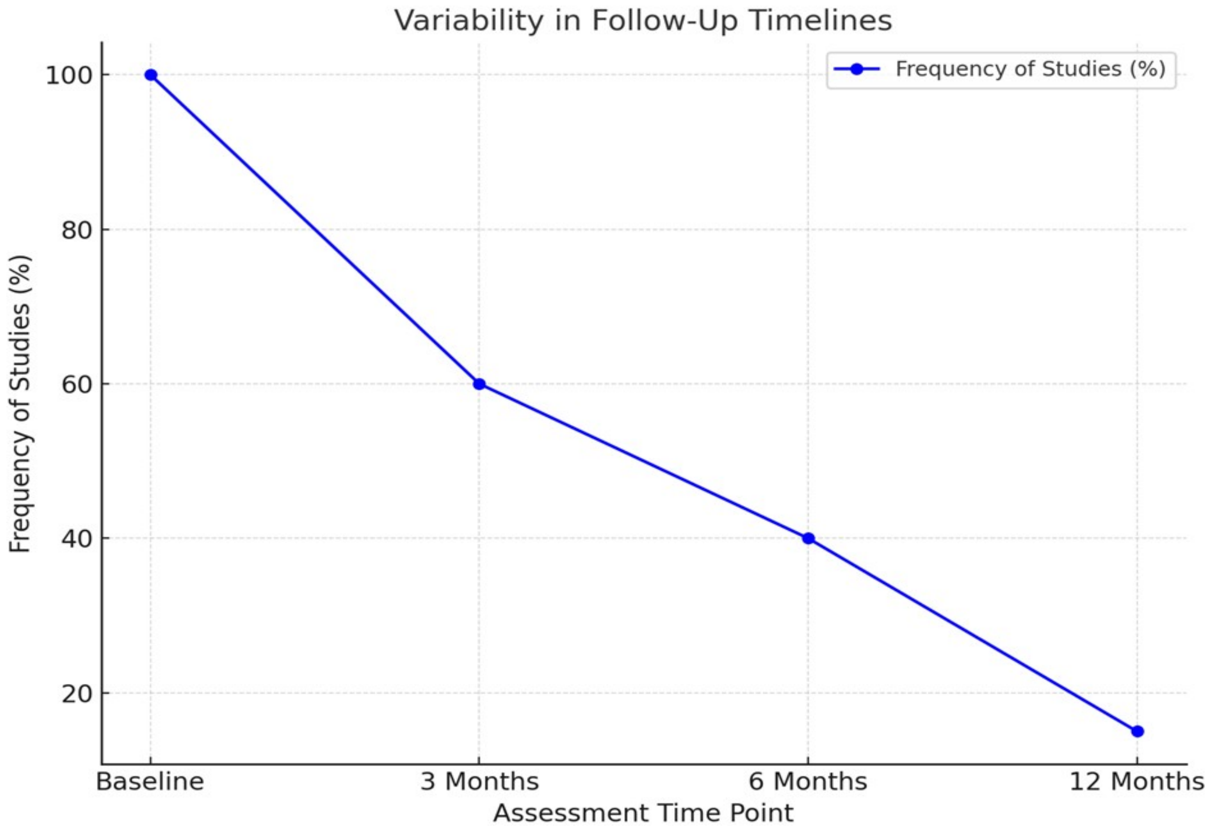


Figure 2: Variability in Follow-Up Timelines Across 69 Studies (2013–2026)

Subtype-Specific Outcomes

The variability in outcome measures across GBS subtypes, such as AIDP, AMAN, and MFS, underscores significant gaps in research methodologies. Addressing these gaps is essential for tailoring interventions and optimizing recovery strategies for patients with diverse clinical and electrophysiological presentations.

AIDP: The Prevalent Subtype

AIDP is the predominant GBS subtype in Western countries, representing 60–80% of cases[29,50]. It is characterized by immune-mediated demyelination, resulting in impaired nerve conduction and progressive muscle weakness. Recovery often hinges on demyelination The MRC scale, commonly used to assess muscle strength, is subjective and demonstrates moderate reliability and validity. Inter-rater reliability scores for the MRC scale range from 0.75 to 0.85, while intra-rater reliability is slightly higher, ranging from 0.80 to 0.90, depending on examiner experience[59]. However, its validity in detecting subtle muscle strength variations is limited, and its sensitivity and consistency are suboptimal for assessing gradual recovery in conditions like GBS and immune-modulating therapies such as IVIg and plasma exchange[24,51]. Saygin et al. found that hand-held dynamometry offers a more precise alternative, with a sensitivity exceeding 85% and high reproducibility compared to the MRC scale[52]. While the MRC scale is practical and widely used, it lacks the precision to detect subtle strength changes, particularly during early recovery phases. Integrating complementary tools, such as hand-held dynamometry (HHD), could enhance sensitivity and provide quantitative measurements[52]. Additionally, the FSS is crucial for evaluating the pervasive and often debilitating fatigue experienced by AIDP patients, which can persist long after motor recovery[28]. Longitudinal studies of AIDP have demonstrated that while early interventions such as IVIg improve acute-phase outcomes, a significant proportion of patients experience residual deficits, including fatigue and sensory impairments. A double-blind RCT by Haridy et al. comparing plasma exchange with IVIg in 81 patients found no statistically significant difference in MRC sum score or GBS Disability Scale at 12 months, confirming functional equivalence over the long term while underscoring the need for extended follow-up beyond the acute phase. For instance, Uz & Karaahmet, observed that 30% of AIDP patients reported persistent fatigue even at 12 months, emphasizing the importance of long-term follow-up and comprehensive outcome assessments[47,62]. The distinct immunopathological features of Guillain-Barré Syndrome subtypes, such as the anti-GM1 and anti-GQ1b antibody associations, reinforce the necessity for subtype-specific outcome assessments. Tools such as nerve conduction studies for AMAN and cranial nerve evaluations for MFS are critical to capturing nuanced recovery trajectories and guiding targeted interventions[11].

AMAN: Challenges and Opportunities

AMAN is more commonly observed in East Asia and Latin America, where it is frequently associated with antecedent infections, such as Campylobacter jejuni[3]. AMAN involves direct immune-mediated damage to motor axons, resulting in rapid and severe motor deficits. Unlike AIDP, Schwann cells are typically spared, facilitating faster axonal regeneration in some patients[42]. Despite its distinct pathophysiology, outcome measures specific to AMAN remain limited. Surve et al., in a decade-long retrospective study of 75 paediatric GBS patients admitted to a neuro-intensive care unit, found that AMAN subtype predominance was associated with worse MRC sum scores at discharge compared to AIDP, yet functional disability assessments beyond standard GBS grading remained sparse in this cohort. Ansari et al. further demonstrated, in a comparative cohort of COVID-19-associated versus non-COVID-19 GBS, that electrophysiological subtyping and MRC sum scores diverged meaningfully between groups, underscoring the need for subtype-sensitive measurement when comparing cohorts. The use of nerve conduction studies (NCS) was reported in only 15% of the reviewed studies, reflecting underutilization of this critical diagnostic and monitoring tool **(Table 3)**. NCS is particularly valuable for AMAN, as it provides objective data on axonal damage and regeneration, which are central to understanding recovery trajectories [53]. Functional mobility assessments, such as the TUG test, were underrepresented in AMAN studies, despite their potential to track improvements in gait and motor function. Integrating mobility and balance assessments could provide a more holistic understanding of functional recovery in AMAN patients. Moreover, studies should evaluate the efficacy of early, targeted rehabilitation programs to maximize axonal repair and prevent secondary complications such as contractures and muscle atrophy.

MFS: A Rare but Distinct Subtype

MFS, characterized by ophthalmoplegia, ataxia, and areflexia, accounts for 5–10% of GBS cases[54,55]. It is strongly associated with anti-GQ1b antibodies, which target gangliosides in cranial nerves[56]. Despite its distinct clinical features, MFS remains underrepresented in GBS research, with few studies employing tools specifically designed to capture its unique manifestations. For example, ocular motility evaluations are essential for assessing ophthalmoplegia but were rarely reported in the reviewed studies. Additionally, ataxia, a hallmark of MFS, could be objectively measured using tools such as the Berg Balance Scale (BBS) or computerized dynamic posturography, but these tools were largely absent from the studies analyzed. This lack of tailored assessments not only limits our understanding of MFS recovery patterns but also hampers the development of targeted interventions.

Table 3: Subtype-Specific Outcome Measures

Subtype	Commonly Used Tools	Frequency (%)
AIDP	MRC Scale, GBS Disability Scale, Fatigue Severity Scale	76
AMAN	Nerve Conduction Studies, MRC Sum Score	19
MFS	Ophthalmoplegia-specific tests	12

Variability and Inconsistencies in Subtype-Specific Research

Heterogeneity in Tools

The choice and application of tools varied widely across studies, even within specific subtypes. For instance, while the MRC scale was a standard for assessing muscle strength in AIDP and AMAN, its grading precision differed between studies, with some employing the full 5-point scale and others using modified versions. This inconsistency reduces the ability to compare results and synthesize findings across cohorts[57]. Similarly, functional mobility tests such as the TUG and FRT were inconsistently applied, with differing definitions of "normal" performance benchmarks[28,45].

Underrepresentation of Psychosocial Outcomes

The review revealed that psychosocial and fatigue outcomes, critical for understanding the broader impact of GBS, were underrepresented across subtypes. For example, while fatigue is a common and debilitating symptom in both AIDP and AMAN, few studies included the FSS in their assessments[37,47]. Additionally, mental health impacts, such as anxiety and depression, were rarely stratified by subtype, despite evidence suggesting that these outcomes may vary based on disease presentation and severity[28].

Cultural and Geographic Disparities

The geographic distribution of studies also influenced the representation of subtypes. AMAN was primarily studied in regions where it is more prevalent, such as East Asia and Latin America, while MFS studies were disproportionately conducted in high-income countries, where diagnostic tools for cranial nerve assessment are more accessible[3]. This geographic bias highlights the need for more inclusive research that reflects the global distribution of GBS subtypes and their outcomes.

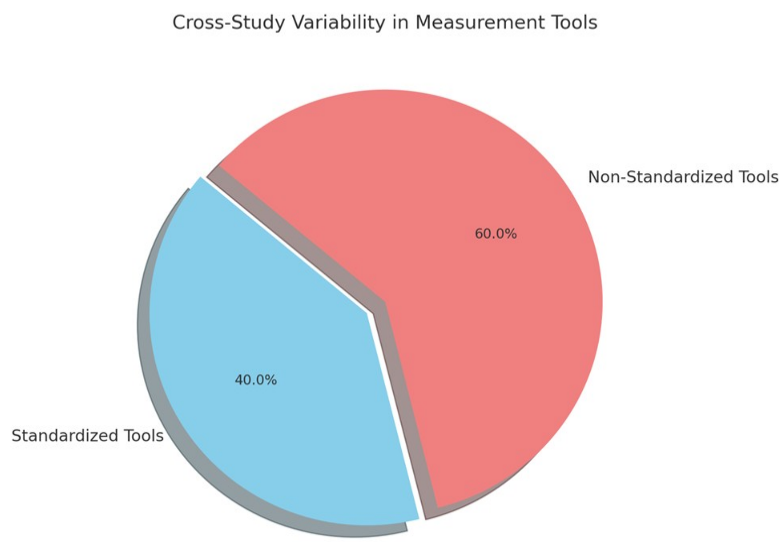


Figure 3: Cross-Study Variability in Measurement Tools

(*Pie chart illustrating the proportion of standardized versus non-standardized tools across 69 studies. Extended search (2024–2026) contributed new tools including GBS-DS, mEGOS, EGRIS, mRS, FIM, serum NfL, PNI, and pain NRS, increasing tool diversity but also heterogeneity*.)

4. Discussion

The findings of this comprehensive review underscore substantial inconsistencies in outcome measurement within GBS research. The primary focus on physical assessments, while clinically significant, reflects an imbalance that overlooks broader psychosocial and functional domains, which are crucial for understanding the full burden of GBS on patients. The variability in assessment tools, timelines, and the underrepresentation of subtype-specific analyses further compound the challenges of synthesizing data across studies.

Physical Assessments and Their Limitations

Physical assessments, particularly muscle strength measurement using the MRC scale, dominated the included studies, with 70% employing this tool. The MRC scale remains a widely used method for quantifying motor strength, given its simplicity and utility for tracking recovery progression in clinical practice[58]. However, despite its ubiquity, the subjectivity inherent in MRC grading reduces its reliability and cross-study comparability[28]. For example, studies have shown that inter-rater reliability in MRC grading can fluctuate significantly, especially among less experienced examiners[59]. Variability in the precision of scoring, particularly in differentiating between Grades 4 and 5, further highlights the need for more objective strength measurements, such as hand-held dynamometry (HHD), which offers higher sensitivity and reliability. The mEGOS demonstrated an AUC of 0.80 for predicting inability to walk independently at six months, while EGRIS achieved sensitivity of 82% and specificity of 70% for identifying patients requiring mechanical ventilation, both validated across independent international cohorts. The pain Numeric Rating Scale (NRS), applied in the largest prospective GBS cohort to date (n=644, Bangladesh), documented pain in up to 89% of patients; its simplicity and cross-cultural applicability position it as an essential but underused PROM in GBS trials[52].

In addition to muscle strength, functional mobility measures such as the TUG and FRT were employed in 45% of the studies. These tools provide valuable insights into balance, gait, and fall risk critical functional domains in GBS recovery[36]. A study by Lee et al. demonstrated that TUG scores correlated strongly with functional independence in GBS patients at six months post-onset[36]. However, the absence of standardized administration protocols across studies undermines the utility of these tools for cross-study comparison. Some studies failed to report normative thresholds for TUG, while others lacked clarity in participant instructions, making data synthesis difficult[28]. Moreover, while physical assessments are critical for quantifying acute-phase motor deficits, they provide an incomplete picture of recovery, particularly in the long term. Residual impairments, such as fatigue and sensory deficits, often persist even after significant motor recovery, underscoring the importance of incorporating additional tools to evaluate these outcomes comprehensively[42].

Psychosocial and Patient-Reported Outcomes

The underrepresentation of psychosocial and patient-reported outcome measures (PROMs) remains a critical shortfall in GBS research. Only 15% of studies assessed psychosocial domains, with tools like the Hospital Anxiety and Depression Scale (HADS) and Short Form-36 (SF-36) being sparsely applied. This limited focus is concerning, as several studies have documented high rates of anxiety, depression, and reduced quality of life (QoL) among GBS survivors, even years after recovery. In a prospective Sri Lankan cohort, Senanayake et al. documented that 42% of patients retained GBS Disability Scores of ≥3 at six months post-discharge and reported significant psychosocial burden, including employment disruption and caregiver dependency, dimensions entirely absent from conventional neurological scoring systems. Furthermore, Lu et al. demonstrated in a propensity-score-matched Chinese cohort that patients receiving structured psychological and behavioural therapy alongside standard rehabilitation achieved significantly higher Functional Independence Measure scores and MRC sum scores at six months compared to standard care alone, reinforcing that psychosocial intervention constitutes a measurable and under-deployed component of GBS recovery[12,28]. Barnes and Herkes, reported that nearly 45% of GBS survivors exhibited clinically significant symptoms of depression, which correlated strongly with reduced physical function and employment rates[23]. Similarly, van den Berg et al. highlighted that GBS patients often experience social isolation and diminished QoL, particularly when psychosocial rehabilitation is not prioritized[42].

Fatigue, a debilitating symptom that persists in approximately 60% of GBS survivors, was also under-evaluated, with only 18% of studies incorporating the FSS[37]. Fatigue significantly impacts functional independence, employment, and social reintegration, yet it remains overlooked in clinical assessments. For example, Walgaard et al. found that persistent fatigue was the strongest predictor of long-term disability and reduced QoL in GBS patients, regardless of motor recovery[45].

The limited use of PROMs such as FSS and SF-36 reflects a broader systemic bias in outcome research, where clinician-centered measures often take precedence over patient-centered assessments. Addressing this gap is essential for ensuring that GBS outcomes align with patients' lived experiences and priorities[40].

Variability in Assessment Timelines

The review revealed significant heterogeneity in the timing of outcome assessments, which poses a major barrier to synthesizing longitudinal data. While most studies conducted baseline evaluations during the acute phase, follow-up timelines varied widely, ranging from three months to one year. Only 15% of studies adhered to consistent assessment intervals, such as evaluations at baseline, 3, 6, and 12 months[28]. This lack of standardization limits the ability to identify recovery trajectories and assess long-term outcomes effectively.

GBS recovery follows a non-linear pattern, often characterized by an initial acute decline, followed by a variable recovery phase that may extend beyond 12 months[3,42]. For example, van den Berg et al. observed that nearly 30% of patients demonstrated continued improvement between 6 and 12 months post-onset, highlighting the importance of prolonged follow-up[42]. However, studies with shorter follow-up periods risk underestimating long-term functional gains and residual deficits, particularly in domains such as fatigue and QoL. Three paediatric studies published between 2024 and 2025 collectively illustrate this gap: Estublier et al. found that 67% of 51 children had persistent sequelae at a median follow-up of 6.3 years, with Fatigue Severity Scale and Multiple Sclerosis Walking Scale scores confirming residual walking limitation even after apparent motor recovery; Surve et al. reported that AMAN subtype predominated in their NIMHANS paediatric neuro-ICU cohort (n=75) and was associated with worse MRC sum scores at discharge compared to AIDP; and Priyadarshini et al. demonstrated in a prospective Indian paediatric PICU cohort that mEGOS and EGRIS scores at admission predicted GBS Disability Score at six months with high accuracy[4,14]. To address this, harmonized assessment timelines—conducting evaluations at baseline, 3, 6, and 12 months are recommended. These standardized intervals allow for robust longitudinal analyses while capturing both early recovery and delayed improvements.

Subtype-Specific Outcomes

Subtype-specific analyses were conducted in only 26% of the reviewed studies, representing a persistent research gap despite modest improvement in the 2024–2026 cohort. GBS comprises distinct subtypes, including AIDP, AMAN, and MFS, each with unique pathophysiological mechanisms and recovery trajectories[3,55]. Studies focusing on AMAN, prevalent in East Asia and South America, often emphasized nerve conduction studies (NCS) to evaluate axonal damage and repair[5]. In contrast, AIDP, the most common subtype in Western populations, has been predominantly assessed using tools like the MRC scale, which may not fully capture axonal regeneration patterns[28]. MFS, characterized by ophthalmoplegia and ataxia, remains underrepresented, with few studies incorporating cranial nerve-specific measures such as ocular motility assessments[7]. Subtype-specific research is essential for tailoring interventions and improving clinical outcomes. For example, axonal subtypes such as AMAN may benefit from early nerve stimulation and physiotherapy targeting axonal repair, while demyelinating subtypes may require prolonged immunotherapy and rehabilitation[55]. Developing subtype-specific tools and incorporating them into a core outcome set will improve the precision and relevance of GBS research.

Critical Analysis of Quality Assessment Scores and Ratings

The manuscript systematically evaluates the quality of 69 studies on GBS using established tools such as the Newcastle-Ottawa Scale (NOS) was applied to observational and cohort studies, the Cochrane Risk of Bias Tool (RoB 2.0) assessed randomized controlled trials (RCTs), and CARE guidelines were used for case reports. The majority of the observational and cohort studies scored between 7 and 9 on the NOS, reflecting high methodological rigor, while most randomized trials demonstrated low risk of bias, indicating robust study designs. Notably, the two highest-quality RCTs identified in the extended search, Haridy et al. comparing long-term plasma exchange versus IVIg outcomes using the MRC sum score and GBS Disability Scale at 12 months, and van Tilburg et al. evaluating serum neurofilament light chain as a longitudinal biomarker, both received low-risk-of-bias ratings, raising the overall evidentiary standard of this review. High-quality studies primarily focused on physical outcomes, subtype-specific recovery patterns, and early-phase interventions. However, studies addressing patient-reported outcomes, psychosocial dimensions, and long-term follow-up often had methodological gaps or smaller sample sizes, leading to moderate quality scores. Additionally, geographic disparities were evident, with research from low- and middle-income countries often underrepresented or of lower quality, although recent contributions from Ethiopia, Bangladesh, Sri Lanka, and Saudi Arabia have begun to narrow this gap. These findings highlight the need for methodological improvements and more equitable research efforts across all domains of GBS outcomes.

Recommendations for Standardization

To address the inconsistencies observed in GBS outcome studies, the implementation of a core outcome set (COS) is strongly recommended. This COS would serve as a standardized framework for selecting tools, domains, and timelines, ensuring consistency and comparability across research efforts.

First, physical outcomes should be systematically assessed using reliable tools. Instruments such as the MRC scale, which measures muscle strength, the TUG test for mobility, and the FRT for balance, are proposed as foundational components of the COS. These tools have demonstrated utility in monitoring critical aspects of physical recovery, yet their consistent application across studies would significantly improve data reliability and comparability.

Second, greater emphasis should be placed on PROMs to capture recovery dimensions that align with patient priorities. Instruments such as the SF-36 for QoL, the HADS for assessing anxiety and depression, and the FSS for fatigue evaluation should be integrated into the COS. These PROMs provide essential insights into the psychosocial and functional burdens of GBS, offering a more holistic perspective on recovery outcomes.

Third, subtype-specific tools must be incorporated to account for the distinct clinical features of GBS subtypes. For example, nerve conduction studies (NCS) are particularly relevant for assessing axonal damage and recovery in AMAN, while cranial nerve evaluations are essential for understanding recovery patterns in MFS. Including such measures will ensure that the COS captures the diverse recovery trajectories of GBS subtypes, promoting precision in both research and clinical interventions.

Finally, the harmonization of assessment timelines is critical for capturing both acute and long-term recovery patterns. A structured schedule, with evaluations conducted at baseline, 3 months, 6 months, and 12 months, is recommended. This approach allows researchers to monitor initial recovery trajectories during the acute phase and identify long-term outcomes, such as residual symptoms or delayed functional gains. Standardizing these timelines will facilitate robust longitudinal analyses and improve the comparability of findings across studies. By implementing a COS that encompasses physical, patient-reported, and subtype-specific measures while adhering to standardized timelines, GBS research can achieve greater consistency and depth. Such a framework will enhance data synthesis, support meta-analyses, and ultimately align research efforts with the needs and priorities of GBS patients.

5. Conclusion

The lack of standardized outcome measures in GBS research presents significant obstacles to advancing clinical knowledge and improving patient care. Variability in the tools, domains, and timelines used to assess outcomes undermines the comparability of data across studies and complicates the synthesis of findings, which are essential for evidence-based clinical decision-making. This inconsistency not only limits the development of universally applicable guidelines but also affects the ability to fully understand and address the multifaceted impact of GBS on patients' physical, functional, and psychosocial health.

The findings of this systematic review emphasize the critical need for a unified framework that integrates robust measures across key domains. The proposed framework should include standardized physical assessments, such as the Medical Research Council (MRC) scale, Timed Up and Go (TUG) test, and Functional Reach Test (FRT), which offer reliable metrics for evaluating muscle strength, mobility, and balance. Equally important is the inclusion of patient-reported outcome measures (PROMs), such as the Short Form-36 (SF-36) for quality of life, the Hospital Anxiety and Depression Scale (HADS) for mental health, and the Fatigue Severity Scale (FSS) for fatigue. These tools are essential for capturing the broader impact of GBS on patient well-being, which often extends beyond physical recovery.

The review also highlights the necessity of integrating subtype-specific metrics into future research. Given the distinct pathophysiological mechanisms and recovery trajectories of GBS subtypes—such as AIDP, AMAN, and MFS—tailored assessment tools, such as nerve conduction studies (NCS) for axonal damage in AMAN or cranial nerve evaluations for MFS, should be prioritized. Such an approach would ensure that outcome measures reflect the nuanced recovery patterns of these subtypes, improving the precision of both clinical assessments and research analyses.

Harmonized timelines for outcome assessments are equally critical. Standardizing evaluations at baseline, 3 months, 6 months, and 12 months would enable researchers and clinicians to capture both the acute and long-term phases of recovery, facilitating robust longitudinal analyses and meaningful comparisons across studies. Consistent timing is particularly important for understanding the progression of residual symptoms and delayed functional improvements, which often vary among patients and subtypes.

Future studies must prioritize the validation of these proposed measures across diverse populations, including underrepresented groups from low- and middle-income countries. Addressing geographic and demographic disparities in GBS research is essential for ensuring that standardized measures are globally applicable and relevant to diverse patient populations.

Additionally, the use of advanced statistical and machine-learning approaches in data analysis could further enhance the interpretation of complex recovery trajectories and the identification of prognostic factors.

In conclusion, the development and implementation of a standardized core outcome set for GBS research is both timely and imperative. By fostering consistency in data collection and analysis, such a framework will facilitate meta-analyses, improve the translation of research findings into clinical practice, and align research priorities with the needs of patients. This endeavor has the potential to significantly advance our understanding of GBS and ultimately improve the care and outcomes of those affected by this debilitating condition.

Competing Interest: Not applicable

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